



Course: CSE207 (Data Structures)

Topic: Final Project

Project Title: Delivery Route Planner Using Dijkstra's Algorithm

Introduction

Efficient delivery route planning is an important real-world application of graph algorithms. A courier service must deliver packages from a **central warehouse** to multiple customers located at different addresses. To minimize delivery time and cost, the courier must find the **shortest possible routes** to each customer through the available roads.

In this project, we model the road network as a **weighted graph**:

- **Nodes (vertices):** represent the warehouse and customers' addresses.
- **Edges:** represent roads connecting these locations.
- **Weights:** represent the distance (or time) required to travel along each road.

By applying **Dijkstra's Algorithm**, we can compute the **minimum distance and path** from the warehouse to every customer in the network.

Objectives

The goals of this project are to:

1. Represent a **delivery network** using suitable graph data structures (adjacency list or adjacency matrix).
2. Implement **Dijkstra's Algorithm** to find the shortest route from the warehouse (node **0**) to each customer node.
3. Output both the **minimum delivery distance** and the **corresponding route (path)** for every customer.
4. Provide a real-world analogy to understand the use of shortest-path algorithms in logistics and delivery systems.

Problem Definition

- The **warehouse** is represented by **node 0**.
- Each **customer address** is represented by a node numbered **1, 2, 3, ..., N**.
- Each **road** between two nodes has an associated **distance**.

The task is to:

1. Read the network input (number of customers, number of roads, and road distances).
2. Apply **Dijkstra's Algorithm** starting from the warehouse (**0**).
3. Print the **shortest distance** and **path** from the warehouse to each customer.

Input and Output Specification

Input Format

1. The first line contains two integers:
 - **N** → number of customers.
 - **E** → number of roads (edges).
2. The next **E** lines each contain three integers:
 - **u v w** → where **u** and **v** are nodes (locations), and **w** is the distance of the road between them.

Output Format

For each customer **i** ($1 \leq i \leq N$):

- Print the **minimum distance** from the warehouse (**0**) to customer **i**.
- Print the **path (sequence of nodes)** representing the route.

Sample Input	Sample Output
4 5 0 1 10 0 2 3 1 2 1 1 3 2 2 3 8	Customer 1: Minimum Distance = 9, Path = 0 → 2 → 1 Customer 2: Minimum Distance = 3, Path = 0 → 2 Customer 3: Minimum Distance = 11, Path = 0 → 2 → 1 → 3 Customer 4: Not reachable